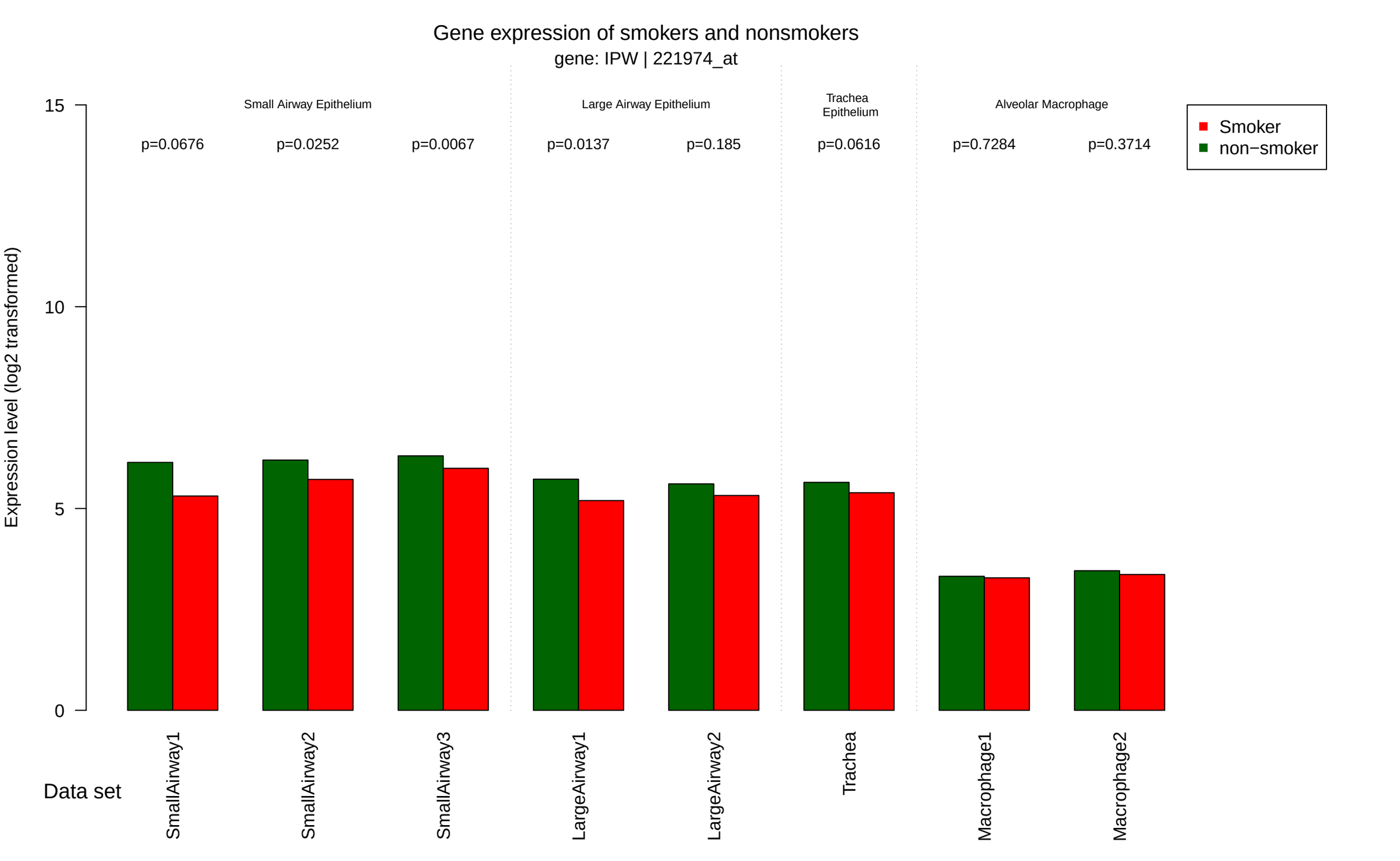
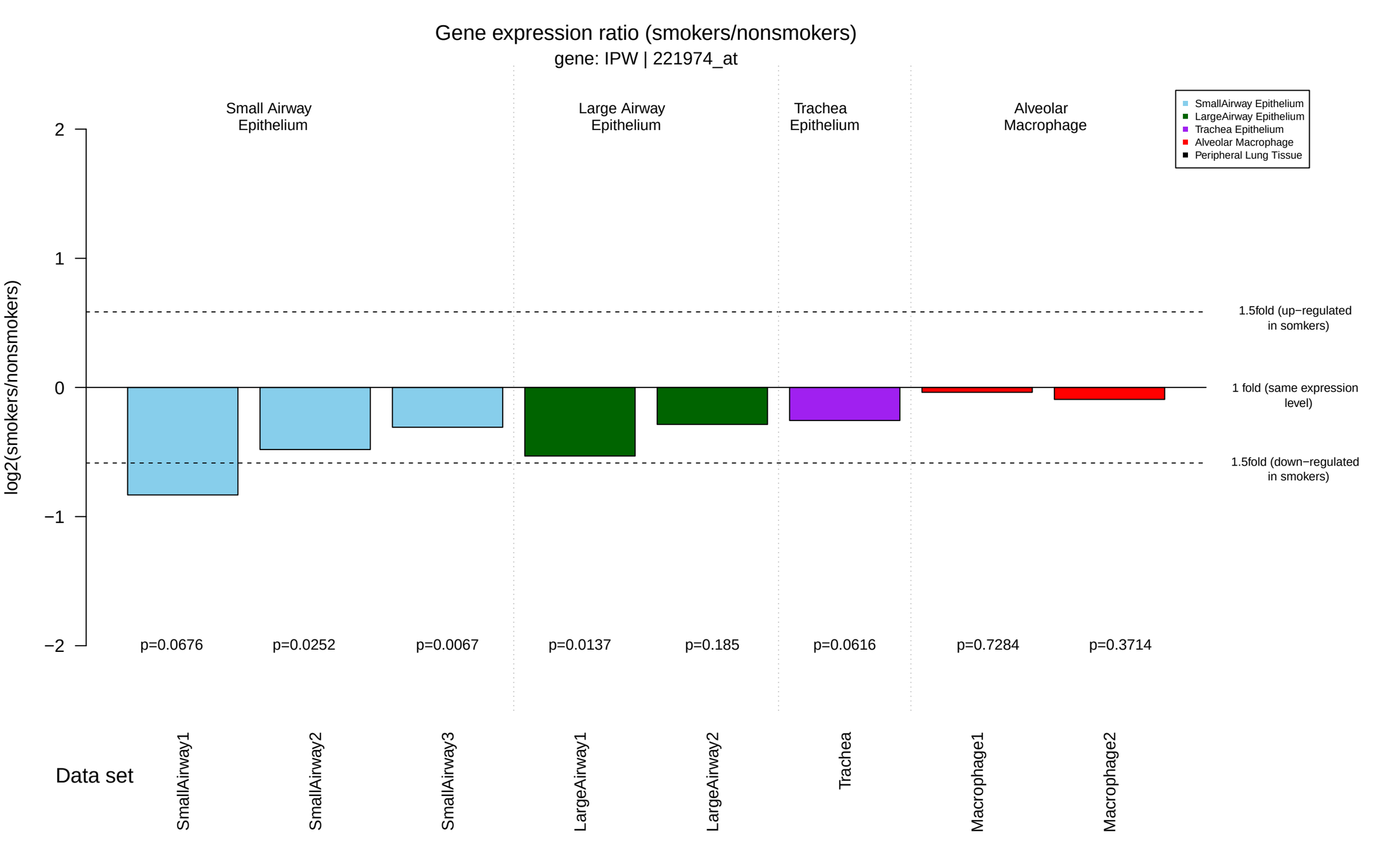
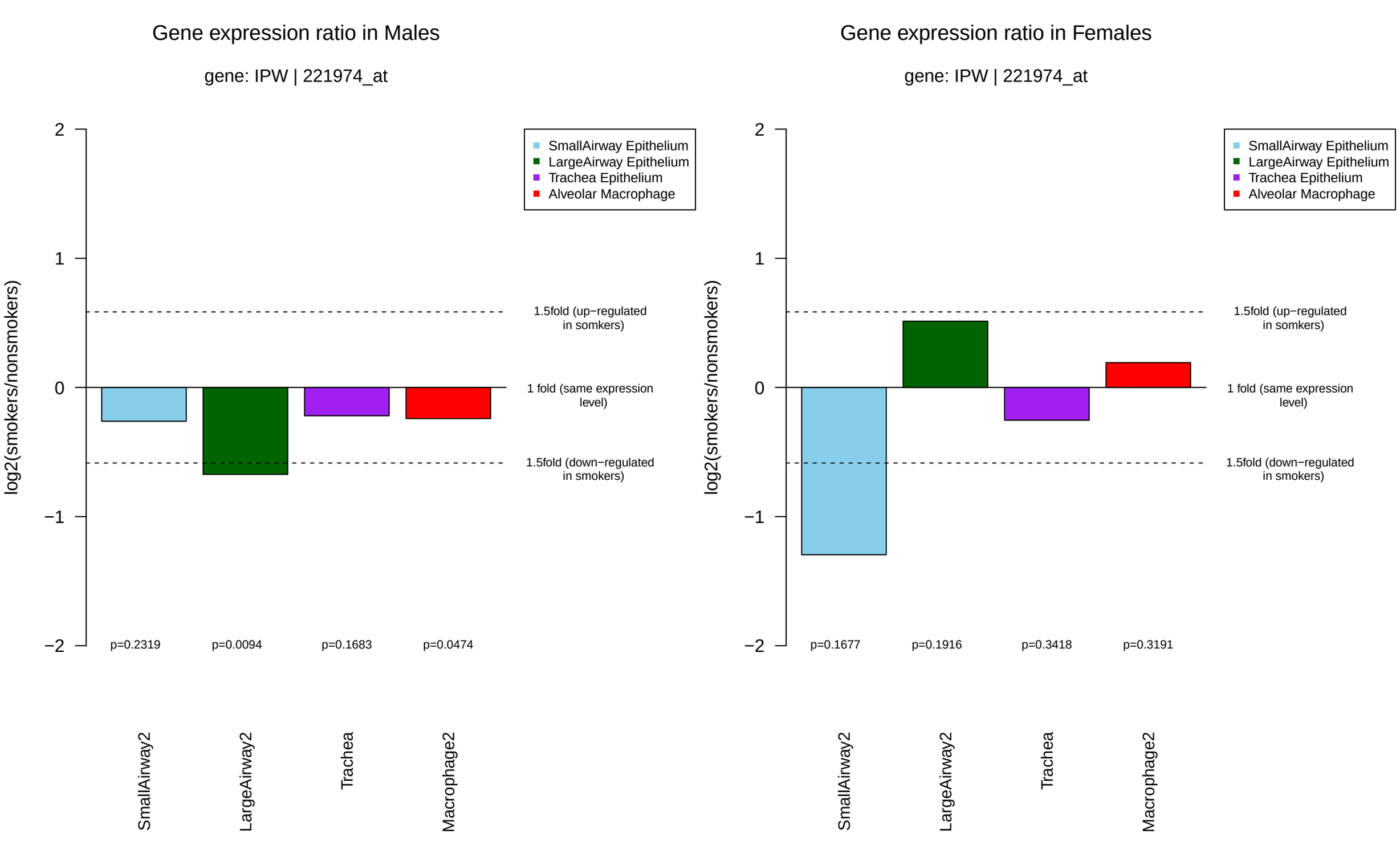
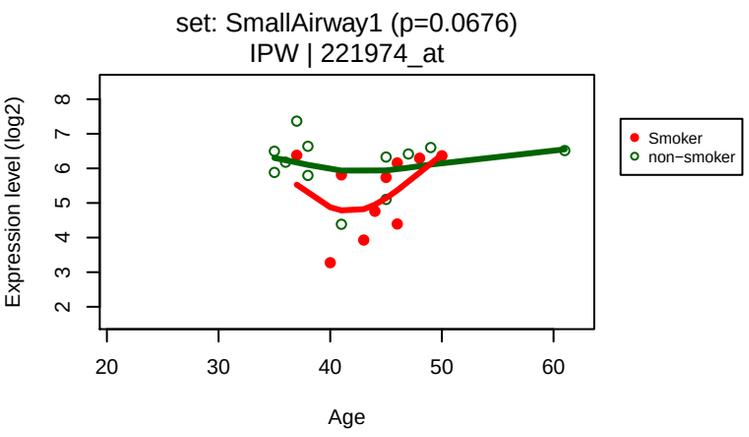
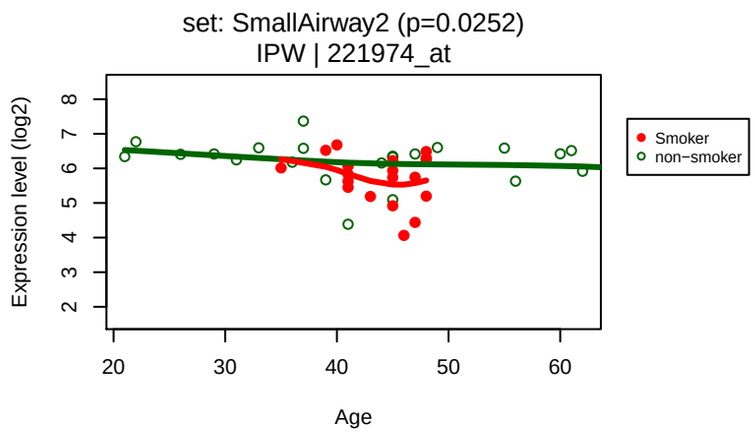
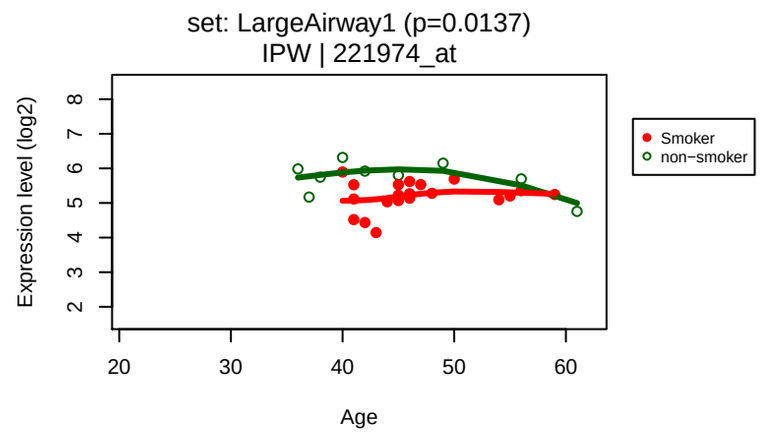

Gene expression of smokers and nonsmokers
gene: IPW | 221974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0676
p=0.0252
p=0.0067
p=0.0137
p=0.185
p=0.0616
p=0.7284
p=0.3714
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

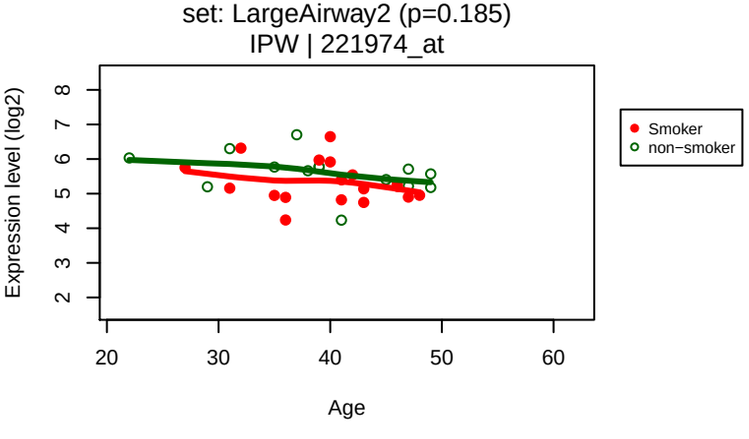

Gene expression ratio (smokers/nonsmokers)
gene: IPW | 221974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0676
p=0.0252
p=0.0067
p=0.0137
p=0.185
p=0.0616
p=0.7284
p=0.3714
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

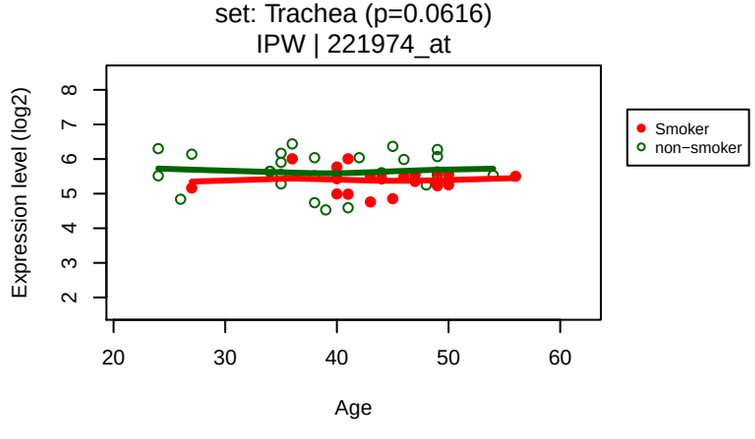

Gene expression ratio in Males
gene: IPW | 221974_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2319
p=0.0094
p=0.1683
p=0.0474
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IPW | 221974_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1677
p=0.1916
p=0.3418
p=0.3191
SmallAirway2
LargeAirway2
Trachea
Macrophage2

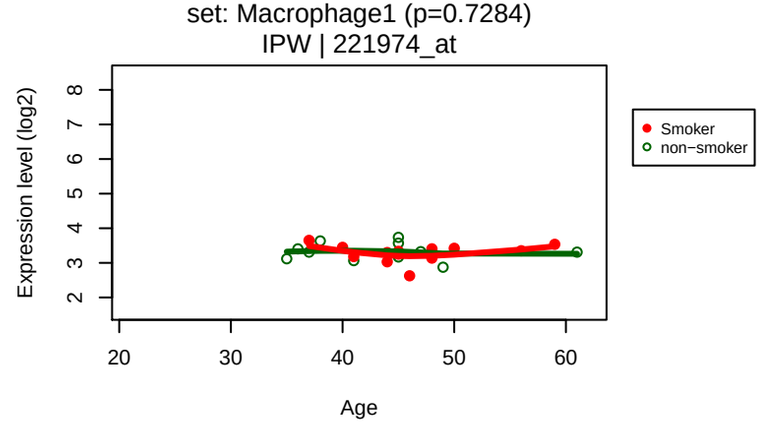

set: SmallAirway1 (p=0.0676)
IPW | 221974_at
Expression level (log2)
Age
Smoker
non−smoker

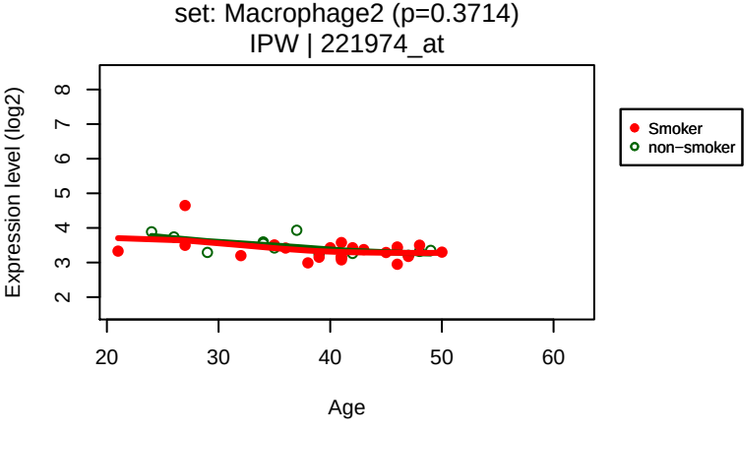

set: SmallAirway2 (p=0.0252)
IPW | 221974_at
Expression level (log2)
Age
Smoker
non−smoker

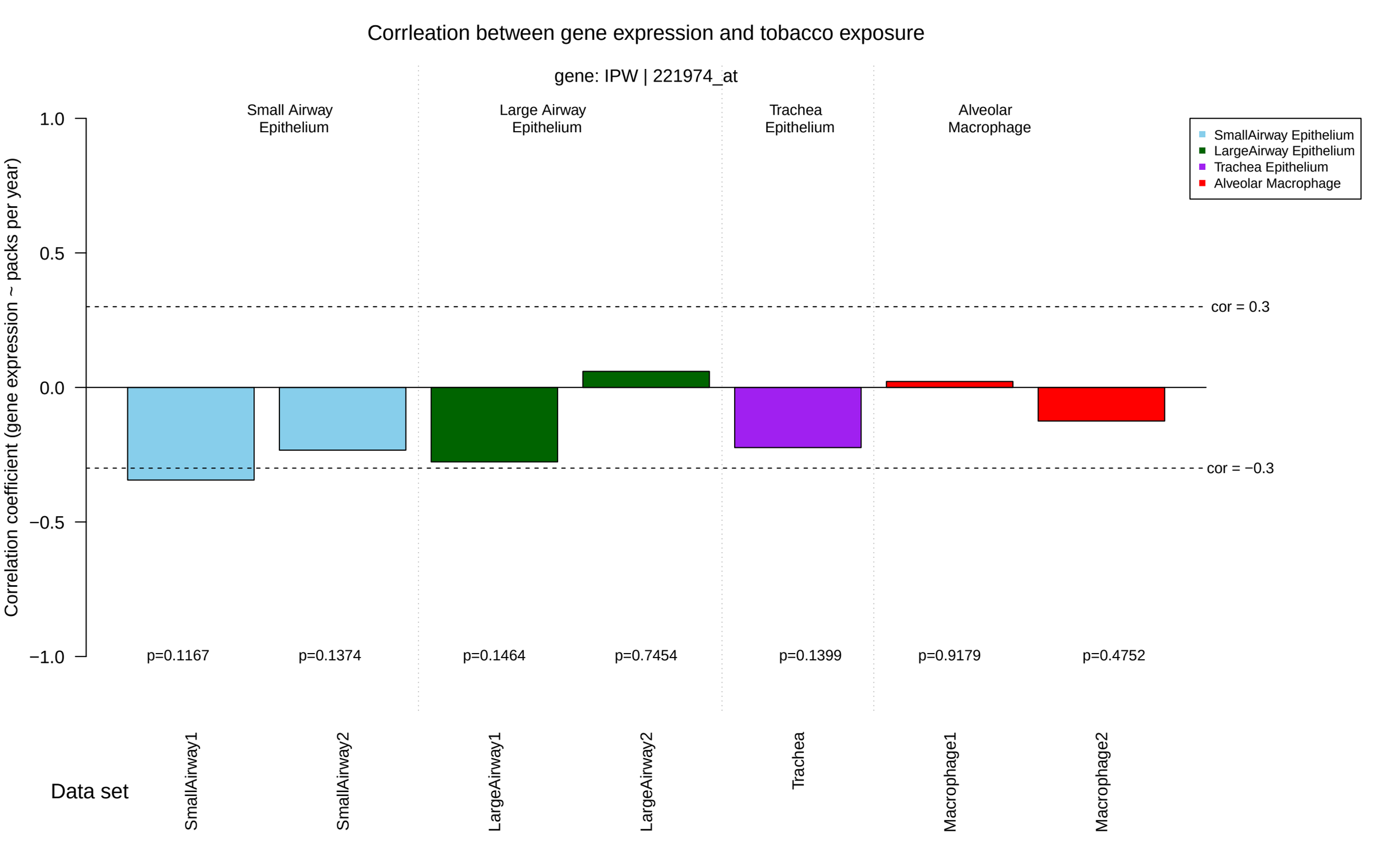

set: LargeAirway1 (p=0.0137)
IPW | 221974_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.185)
IPW | 221974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0616)
IPW | 221974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7284)
IPW | 221974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3714)
IPW | 221974_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IPW | 221974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1167
p=0.1374
p=0.1464
p=0.7454
p=0.1399
p=0.9179
p=0.4752
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2